

## 3.2 Conditional Probability and the Multiplication Rule

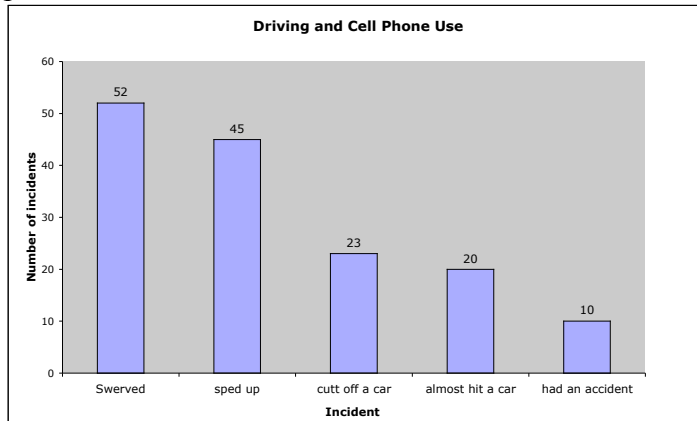
1. Three people are selected at random. Find the probability:
  - a) all 3 are born on Tuesday
  - b) all 3 are born on the same day of the week
  - c) all 3 are born on a different day of the week
2. A 3 person jury must be selected at random from a pool of 12 people that has 7 men and 5 women. Find the probability of selecting an a) all-female jury, b) all-male jury.
3. You purchase 20 lottery tickets for which the probability of winning some prize on a single ticket is 1 in 10. What is the probability that you have at least 1 winning ticket among the 20 tickets?
4. Find the probability of drawing 3 diamonds in a row from a regular deck of cards
  - a) the drawn card is returned to the deck each time
  - b) the drawn card is not returned to the deck each time
5. Refer to problem #4. Find the probabilities if you are drawing 3 fives in a row.
6. Of campus professors 60% are male, and of these, 15% work for College of Humanities. Find the following probabilities:
  - a) randomly selected professor is a male and works for College of Humanities.
  - b) randomly selected professor is a male and does not work for College of Humanities.
  - c) randomly selected professor is a female and works for College of Humanities.
7. 

| Student ages | Frequency |
|--------------|-----------|
| 17 – 26      | 149       |
| 27 – 36      | 85        |
| 37 – 46      | 46        |
| 47 – 56      | 15        |
| 57 and over  | 5         |

  - a) Find the probability that a student chosen at random is between 27 and 36 years old.
  - b) If 4 students are randomly selected (without replacement), find the probability that all four students are between 37 and 46 years old.
  - c) If four students are randomly selected (without replacement), what is the probability that at least 1 will be 57 years or older?
  - d) If four students are randomly selected (without replacement), what is the probability that none of these four students are between 17 and 26 years old?
8. The probability that a person in the U.S. has type A+ blood is 32.5%. Five unrelated people in the U.S. are selected at random. Find the probability that:
  - a) all five have type A+.

c) at least one of the five has type A+.

9. The following graph shows the types of incidents encountered with drivers using cell phones.



- Find the probability that a randomly chosen incident involves cutting off a car.
- Find the probability that two randomly chosen incidents (without replacement) both had an accident.
- Find the probability that a randomly chosen incident did not involve cutting off a car.

10. The access code for a garage door consists of 5 digits. The first digit can not be 0. Find the probability that you guess the code from the first try?

### 3.3 The Addition Rule

- Find the probability of rolling either a 2 or a 5 when you roll a single die.
- Find the probability of drawing either a queen or a spade from a standard deck of cards.
- You select one person at random from a room with 15 people: 5 blond men, 2 dark hair men, 4 blond women and 4 dark hair women. What is the probability that you will select either a women or a blond person?
- If you roll a die, find the probability that you roll a number greater than 4 or an odd number?
- A card is selected at random from a standard deck of cards. Find each probability:
  - Randomly selected card is a black suit or a king.
  - Randomly selected card is a four or a face card.
  - Randomly selected card is a 7 or a king.

6.

|         | Psychology major | Non-Psychology major |
|---------|------------------|----------------------|
| Males   | 150              | 9,750                |
| Females | 375              | 11,300               |

A student is selected at random. Find the following probabilities:

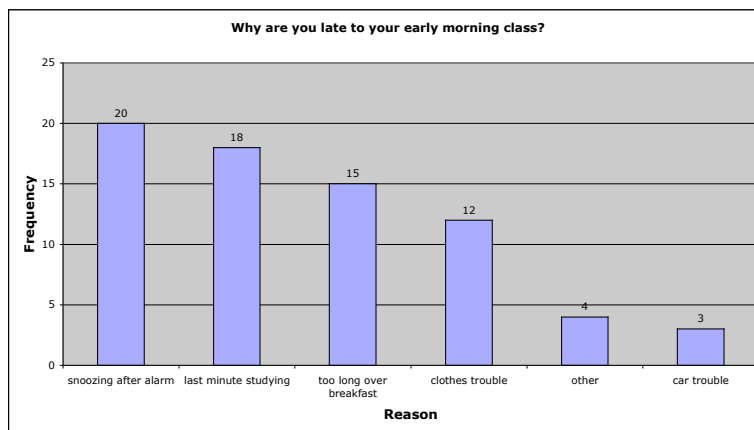
- b) the student is a male or not psychology major
- c) the student is a male and psychology major
- d) Given that the student is a female, what is the probability that she is a psychology major?
- e) If we select 3 students at random (without replacement), what is the probability that all 3 students are women and majoring in psychology?

7. Sam is playing a game with 2 dice (black and red) and on his next move he needs to throw a sum bigger than 9 in order to win. What is the probability that Sam will win?

8. If you roll a die, find the probability that you roll a number less than 5 or an even number.

9. If we have 3 sets A, B and C, find  $P(A \text{ or } B \text{ or } C)$ .

10. From Section 2.2. we had this Pareto chart:



If you randomly selected a person from a sample, find each probability:

- a) The person is late because of last minute studying or clothes trouble.
- b) The person is not late because of last minute studying.

If you randomly selected 4 people from the study (without replacement), what is the probability that all 4 were late because of car trouble?

If you randomly selected 4 people from the study (without replacement), what is the probability that all 4 were late because of trouble with clothes?

### 3.4 Additional Topics in Probability and Counting

1. In how many different ways can we arrange letters A, B, C, D ?
2. We have 4 objects, A, B, C, and D, and we want to make ordered arrangements of 2 objects. How many would we have?

3. How many distinguishable permutations for MATHEMATICS?

4. Suppose you coach a team of 12 swimmers and you need to put together a 4-person relay team. In how many different ways can you do this?

5. Calculate:

|                |                                   |                                 |
|----------------|-----------------------------------|---------------------------------|
| $15! =$        | ${}_{25}P_3 =$                    | $\frac{{}_{10}C_3}{{}_{40}C_3}$ |
| $12! =$        | ${}_{40}C_3 =$                    | $\frac{15!}{12!} =$             |
| ${}_{10}C_3 =$ | $\frac{{}_{40}P_3}{{}_{35}P_7} =$ | $\frac{18!}{13!} =$             |

6. The scholarship committee is considering 25 applicants for 3 awards (1<sup>st</sup> award - \$3,500, 2<sup>nd</sup> award - \$3,000, 3<sup>rd</sup> award - \$2,000).

How many different ways are possible to award these scholarships?

7. There are 30 passengers that still need to check-in and get a boarding pass. The airline representative will upgrade 5 passengers to the first class, seats 1B, 1D, 3A, 3C, 4B. In how many different ways can the airline representative do this?

8. 20 runners enter the competition. In how many ways can they finish 1<sup>st</sup>, 2<sup>nd</sup>, and 3<sup>rd</sup>?

9. How many ways can 3 Republicans, 2 Democrats, and 1 Independent be chosen from 10 Republicans, 8 Democrats, and 5 Independents to fill 6 positions on City Council?

10. A security code consists of 2 letters followed by 3 digits. The first letter can not be A, B, or C, and the last digit can not be a 0. What is the probability of guessing the security code in one trial? 2 trials?

11. A shipment of 40 fancy calculators contains 5 defective units. In how many ways can a college bookstore buy 20 of these units and receive:

a) no defective units

b) one defective unit

c) at least 17 good units

d) What is the probability of the bookstore receiving 2 defective units?

12. You are dealt a hand of four cards from a standard deck. Find the probability that:

a) the first three cards are of the same suit and one is of a different suit.

b) three cards are of the same suit and one is of a different suit